LOSSFUNK
Auto-Research Stage

Research Question - *Why do World Models face roll-out drift?*

Roll-out Drift (World Models)

- **Definition:** The cumulative error that occurs when a model's long-term predictions gradually diverge from physical laws or reality.
- **The Cause:** Small errors in initial prediction steps compound over time, causing the simulation to "drift" into logically or physically impossible states.
- **The Result:** The model maintains high internal coherence (it looks realistic) but loses factual or causal accuracy (it violates the rules of the world).

Github Link : https://github.com/alwayshubh2595/Lossfunk_Submission_Shubh_Srivastava

Shubh Srivastava

- My original framing in the Research Question , treated rollout drift as a regularizer problem. In my June 17 lossfunk call, Ayush raised the possibility that drift might have multiple causes other than just the regularizer. I found the lead worth investigating and made it the question this submission set out to answer.
- The series of experiments try to find out experimentally the causes behind roll out drift in world models .

k	KL_sigreg	Ak_sigreg	KL_noreg	Ak_noreg
1	1.48	1.13	0.43	1.02
5	0.75	1.29	0.66	0.81
10	0.15	2.00	1.28	0.87
20	1.01	4.40	4.15	0.99
50	17.68	18.02	87.05	2.78

(my previous experiment's result , showing roll-out drift)

Required Mathematical Background for understanding the following experiments

- **KL Divergence** - Used to measure how close two probability distributions are , a lower score indicates similarity
- **Amplification Ratio (A_k)** - Given two vectors have very small deviation at t=0 , it measures how far do the two vectors drift after after a certain t = t , it is used to measure how far apart the the initial input is compared to the same input with some perturbation .
- **SIGReg** - A regularizer that forces the latent space to be in an Isotropic Gaussian distribution
- **Lyapunov Exponents** - Quantify the average exponential rate at which infinitesimally close trajectories in a dynamical system diverge or converge.

Autoresearch Exp 1 : Training Protocol Dominates Rollout Drift in Latent World Models: A Causal Attribution Study

System used - Autovoila

Changes made to the original system -

The first time I used the Autovoila system on a different project without any changes I realized that if i am already working on a topic and not like a complete beginner , that means i have some idea of what / how i wanna do my research , but the autoresearch system just by taking in the topic / prompt after the “read voila.md” can't actually judge what's in my mind .

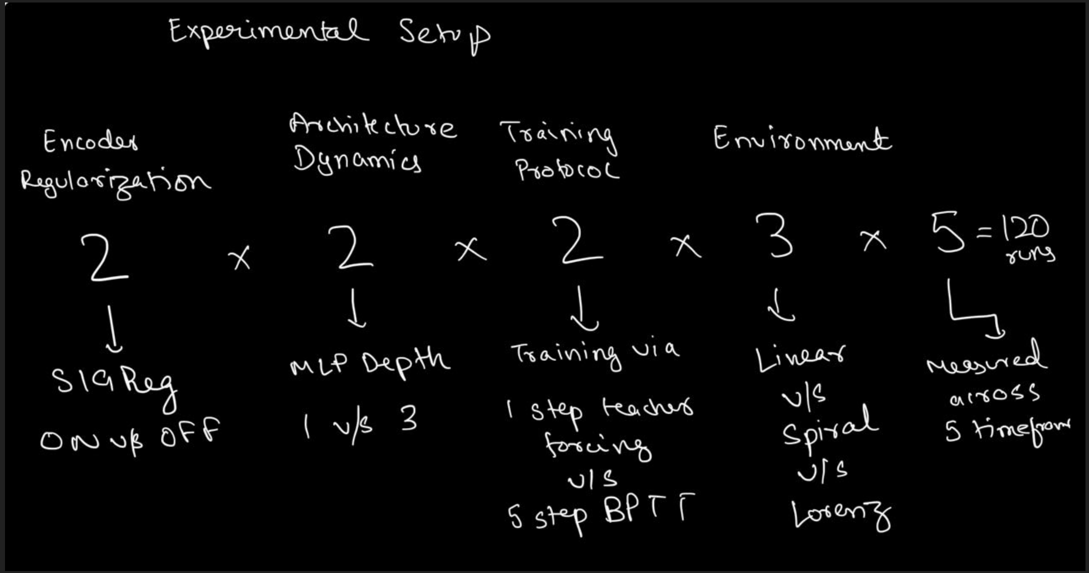
To get more control over the run - I created a new file preliminary_experiment_brief.md , which had some experimental details mentioning what all I need in the final output in detail and a few ways in my mind to approach it . It also had the summary of my previous findings, and the required changes were also done on voila.md . **In this run , I already had a definitive research question in my mind and also had an experimental process for it** , hence less experimental and planning autonomy was given to the system

Main Research Question - Trying to find out what all components might be responsible behind the roll-out drift caused in world models ?

Paper Link - https://github.com/alwayshubh2595/Lossfunk_Submission_Shubh_Srivastava/blob/main/Paper%201.pdf

Research Question , Experimental Setup and Results (in brief) -

We test this claim directly via causal attribution. We ask: of the total rollout amplification A_k observed in latent world models, what fraction of variance is attributable to (a) encoder regularization, (b) dynamics capacity, and (c) training protocol?



Results :-

The causal attribution experiment reveals that the <u>training protocol (1-step teacher-forcing vs. 5-step BPTT) is the dominant factor in rollout drift, significantly outweighing encoder regularization.</u> Variance decomposition of $\log A_{50}$ shows that the training protocol accounts for 63.6% of the variance in stable-linear environments and 43.7% in the Lorenz system.

A few positives from the paper

Factorial structure. We run a $2 \times 2 \times 2$ design: F1 (SIGReg on/off) $\times$ F2 (dynamics MLP depth: 1 vs. 3 hidden layers) $\times$ F3 (training: 1-step teacher-forcing vs. 5-step BPTT), giving 8 conditions.

Environments. (i) *Stable-linear*: $\mathbf{z}_{t+1} = A\mathbf{z}_t$, eigenvalues $\{0.7, 0.8, 0.9\}$, ground-truth $\lambda_{\max} < 0$. This is the sanity check: A_k must shrink. (ii) *2D spiral*: rotation + 5% expansion per step, $\lambda_{\max} \approx +0.049$. (iii) *Lorenz*: σ=10, ρ=28, β=8/3, dt=0.01, $\lambda_{\max} \approx +0.906$.

For each environment, 50,000 trajectories of length 100 are generated (80/10/10 split). 5 seeds per condition, 150 total runs. All models train for 20,000 steps, batch size 8,192, AdamW (lr $= 3 \times 10^{-4}$, wd $= 10^{-4}$). Latent dimension $d_z = 16$. Encoder: 2-layer MLP (width 256, GELU, LayerNorm).

1. The experimental setup is robust and and clear , the addition of a Sanity Check environment (stable linear) environment helps build more trust on the result , because that is the one for which we are already sure of the result before the experiment

All the training and run details are mentioned clearly and is largely accurate (except one thing, mentioned in the criticism)

2. the noise addition in the input data is carried out via 3 different methods and results are analyzed independently , eliminating scope for any bias .
Even in real world model input data , the perturbations are never in the same format , hence this is a bit closer on how we can get simulate to real world data while still using synthetic data.

Rollout amplification. We measure A_k via three complementary methods. The perturbation method computes:

$$A_k^{\text{pert}}(\varepsilon) = \mathbb{E}_{\mathbf{z}_0,\mathbf{u}}\left[\frac{\|f^k(\mathbf{z}_0 + \varepsilon\mathbf{u}) - f^k(\mathbf{z}_0)\|}{\varepsilon}\right] \tag{1}$$

where $\mathbf{u}$ is a random unit vector and $\varepsilon = 10^{-3}$. The Jacobian method computes the operator norm of the accumulated product Jacobian $\prod_{i=0}^{k-1} \partial f/\partial \mathbf{z}|_{\mathbf{z}_i}$. The trajectory-divergence method measures how the spread of nearby initial conditions evolves.

3. After the results are analysed , agreement and diagnostic checks mathematically confirm that the experiment actually successful in carrying out the required

4.3 Method Agreement

Perturbation and trajectory-divergence methods agree closely (within 20% for most conditions). The Jacobian method gives higher values at $k \leq 10$ (computed via exact Jacobian products) and is consistent in direction with perturbation. The Jacobian method is not computed at $k > 10$ (flagged deviation; numerically near-singular).

4.5 Diagnostics

SIGReg successfully drives marginal KL to near-zero (1–53 nats vs. 37–969 for NoReg).

My criticisms of the paper :-

A popular diagnosis [Anonymous, 2025] attributes drift to marginal regularizers such as SIGReg, which constrain the distribution $q(\mathbf{z})$ aggregated over timesteps but place no constraint on the conditional $p(\mathbf{z}_{t+1}|\mathbf{z}_t)$. The implicit claim is: *fix the regularizer and rollout drift improves.*

this specific citation refers to - LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels paper by AMI Labs . Nowhere in the paper is it claimed explicitly that the cause of drift is the regularizer (SIGReg) and it doesn't even recommend clearly a fix for the regularizer. The autoreseatch system hallucinated and presented my previous hypothesis as an already existing claim . The paper sets up a version of my hypothesis that's easier to refute than what I actually claimed, then refutes that easier version.

Marginal regularizers. SIGReg [Anonymous, 2025] pushes the empirical marginal $q(\mathbf{z})$ toward $\mathcal{N}(0, I)$ via a characteristic-function distance. We use a moment-matching proxy: $\mathcal{L}_{\text{reg}} = \|\boldsymbol{\mu}_{\text{batch}}\|^2 + \|\boldsymbol{\Sigma}_{\text{batch}} - I\|_F^2$.

A proxy of the actual SIGReg regularizer was used instead of the real function and no justification was given for this specific design choice in the entire paper. This leads to a very easy alternative explanation that the results observed in this paper might never actually transfer to the real SIGReg paper introduced in the LeWM paper. However the line “SIGReg successfully drives marginal KL to near-zero” written in a later section , shows that the even the proxy used did it’s job well , but it still makes the whole point of the paper a bit weaker in my opinion

My criticisms of the paper (contd) :-

The entire paper uses a lot of mathematical concepts in the experimental setup as well as in the results section

Eg. Table 1 shows the variance decomposition of $\log A_{50}^{\text{pert}}$ per environment.

Table 1: Variance decomposition of $\log A_{50}^{\text{pert}}$ (η^2, fraction of total variance). F1=SIGReg, F2=dynamics capacity, F3=training protocol.

Environment	F1	F2	F3	F1×F2	F1×F3	F2×F3	Residual
Stable-linear	0.126	0.052	**0.636**	0.035	0.039	0.027	0.085
Spiral-2D	0.106	**0.251**	0.127	0.132	0.014	0.033	0.337
Lorenz	0.277	0.024	**0.437**	0.015	0.138	0.004	0.105

As a reader , who already has descent understanding of all the background mathematical concepts , it was really tough for me to understand the results and justifications behind them . I got this same problem while working on a different problem with auto-research , hence the mathematical backing and the simplicity is missing in my opinion .

Environments. (i) *Stable-linear*: $\mathbf{z}_{t+1} = A\mathbf{z}_t$, eigenvalues $\{0.7, 0.8, 0.9\}$, ground-truth $\lambda_{\max} < 0$. This is the sanity check: A_k must shrink. (ii) *2D spiral*: rotation + 5% expansion per step, $\lambda_{\max} \approx +0.049$. (iii) *Lorenz*: σ=10, ρ=28, β=8/3, dt=0.01, $\lambda_{\max} \approx +0.906$.

For each environment, 50,000 trajectories of length 100 are generated (80/10/10 split). 5 seeds per condition, 150 total runs. All models train for 20,000 steps, batch size 8,192, AdamW (lr $= 3 \times 10^{-4}$, wd $= 10^{-4}$). Latent dimension $d_z = 16$. Encoder: 2-layer MLP (width 256, GELU, LayerNorm).

Side sweep. Latent dimension $d_z \in \{2, 8, 32\}$ on the 2D spiral (SIGReg + small + 1-step).

Oracle baseline. Dynamics trained directly on ground-truth states (large + 5-step BPTT).

As mentioned in the experimental setup section - the total experiments account to 2x2x2x3x5 = 120 runs, however it is mentioned as 150 here . This is because the balance 30 belong the Side Sweep and Oracle Baseline section. The reason behind keeping them is not explained clearly in the paper and even in the results section there is no direct mention of them , directly leading to confusion .

My criticisms of the paper (contd) :-

4.2 Sanity Check Failure

Critical finding: 17 of 40 stable-linear conditions exhibit $A_{50}^{\text{pert}} > 1$. For 1-step-trained models, A_{50} reaches values of 131 (NoReg_Small_1step), 952 (SIGReg_Small_1step), and 266 (SIGReg_Large_1step) on average across seeds. This is impossible if the latent dynamics faithfully represent the underlying linear system.

The explanation lies in the encoder geometry: 1-step teacher-forcing trains the encoder and dynamics jointly but only ever supervises single-step transitions. The encoder is free to warp the latent space in ways that preserve 1-step prediction accuracy while creating geometrically unstable structure (e.g., expanding directions that cancel over one step but compound over many steps). SIGReg enforces a Gaussian marginal, which forces the encoder to use more of the latent space (effective $d_z \approx$ 9–10 vs. 2–5 for NoReg conditions; see Appendix A). A fuller latent space under 1-step training creates more opportunity for geometric instability.

This section refers to the the stable linear environment that was added as a sanity check , because we for sure knew that A50 would be <1 , and the same would confirm the correctness of the setup . However , the results show that in 17 / 40 cases it's actually > 1 .

The explanation tied to it is that , since its only predicting 1 step , it makes the encoder free and gives more space to SIGReg to to enforce the isotropic Gaussian . All this leads to a geometrical instability . However , why is it true for only 17 cases ? , and even the explanation given in not very convincing , hence its something that AI just wrote to create an explanation and not the true reason

Auto - Research System

After making the suitable changes in the autovoila system (mentioned on page 5) which helped me set the right scope and focus for my research question, I still feel there are a few limitations in the current system.

1. **Compute Estimation Problem** - I have observed that claude code often overestimates/underestimates the scope of the experiment based on the given compute . While planning I had assigned a 5$ worth of GPU compute to it and in the middle of the run the system realized that the planned experiment is too big to be completed in this timeframe and it reiterated and planned a new 3 hour experiment which actually on running got completed within 1.5 hours . This lack of planning leads to waste of resources and confusion on the actual scope of the project .
2. **Mathematical Background -** A previous run on auto-voila on the topic “Reinforcement Learning in Hyperbolic Spaces” as well as this current run made me realize that the LLMs have a tendency to not include a lot of mathematical intuitions behind the given procedures and results
3. **Scope and Focus** - The original auto-voila system works really well if I want a free run on an entirely new topic , however if I have some ideas on the project its tough to convey it to claude code via one line in CLI , hence I added a .md file in this run .

What I learnt about my Research Question?

Even with the flaws like a proxy regularizer among other things , I felt convinced enough that rollout drift is not only a regularizer problem; it is also shaped by the training protocol and the environment the model is trained on. That expands the scope of my original hypothesis considerably.

Even logically, training protocol determines what data distribution the model actually sees during optimization — one-step teacher forcing never exposes the model to its own compounding errors, while multi-step rollout training does — so it would be surprising if protocol had no effect on drift at all. The first autoresearch run made this concrete rather than theoretical: it gave me a specific mechanism (encoder geometry under teacher forcing) and specific numbers to reason about, even though the mechanism itself remained unproven.

Paper Link - https://github.com/alwayshubh2595/Lossfunk_Submission_Shubh_Srivastava/blob/main/Paper%202.pdf

Autoresearch Exp 2 :

System used - Auto Voila

When Collapse Censors Drift: Differential Encoder Collapse Pre-empts Regularizer–Protocol Attribution in Latent World Models

(A failed / negative result experiment)

Changes made to the original system -

To solve the compute miscalculation problems a specific line in voila .md was added

```
- you would be given compute access after the experiment is planned - I have 30$ credit on jarvis lab ai , you have to
let me know which GPUs you would need and for how many hours . $30 is the hard ceiling for this scenario. Design
accordingly. If the best design requires more, tell me the tradeoffs — don't silently scale down without flagging it ,
but treat it as a hard ceiling that you have to fully utilise do a rigourous calculation of the compute before hand ,
you are free to use use multiple gpus if you wish and require ,   you should not realize in the middle of the
experiment that the given compute is too little or too much for the planned experiment and then when everything is
decided ask me to give you compute access via ssh using jarvis labs.ai
```

To increase the maths component in the paper I added a line in voila.md

```
- Ensure that all the maths used in the paper is well explained , work on the mathematical clairty of this paper
well . The mathematical correctness is directly propotional trust
- In your design, explicitly state which regularizers you're using and whether you're using their canonical
formulations (e.g., real characteristic-function SIGReg, not a moment-matching proxy) or approximations. If you're
using approximations, justify the substitution or acknowledge that findings will need caveating accordingly.
```

An important note - The first time I attempted this experiment the run failed midway due to some mistake on my end , but i had the logs and planning of GPT 5.5 (extra high effort) saved even though the run did not complete - I added those files in this run as a reference material that it could access for borrowing ideas or building more work on .

Core Research Question

This run builds upon the previous autoresearch run , using a real implementation of SIGReg instead of a proxy one and increases the scale of the experiment with a higher compute given to the autoresearch run . The papers checks whether whether a faithful version of SIGReg (a real Gaussian-normality regularizer for latent world models) aligns with our prior study's conclusion .

Experimental Setup

a) Common Part : - an encoder (2-layer MLP, width 128, d=16 for state observations; 4-block CNN, d=32 for pixels) and a transition MLP over [z, a]. Just like the LeWM paper , there is no decoder, reward head, stop-gradient, or EMA target with prediction plus regularization being the only losses

b) 3 cases for Regularizers :

R0 - No Regularizer | R1 - a proxy SIGReg (like the previous run) | R2 - a faithful implementation of SIGReg based on the Epps–Pulley statistic

c) 3 cases for Training protocol :

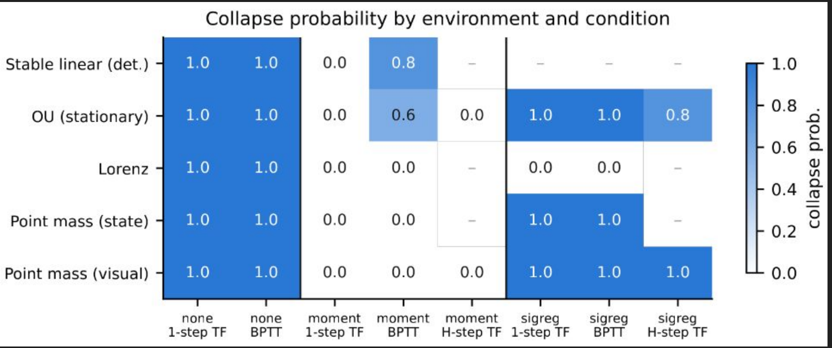
P0 (1-step teacher forcing): always predict one step ahead, always starting from the real previous state
P1 (H-step closed-loop, H=8): predict 8 steps ahead, feeding its own guesses back into itself
P0.5 (H-step teacher forcing, control): predict 8 steps ahead too, but each step still starts from the real previous state — this isolates "more training signal" from "practicing on your own mistakes"

d) Environments:

3.3 Environments

E0-det (bridge only): deterministic stable linear, $x_{t+1} = Ax_t$, eigenvalues $\{0.7, 0.8, 0.9\}$; the prior study's sanity check, retained for R0/R1 bridge cells. **E0 (stationary OU):** same contraction with noise calibrated so the marginal is $\mathcal{N}(0, I_3)$ at every step; SIGReg's favorable regime by construction, added because a time-varying marginal makes the Gaussian target unfair to it. **E1 (Lorenz):** σ=10, ρ=28, β=8/3, dt=0.01; chaotic, so drift must be measured as excess over ground truth. **E2/E3 (action-conditioned point mass, state/pixels):** identical clipped dynamics; E3 renders 64×64 grayscale frames, isolating the pixel-encoder axis.

Main Results



Each box shows the collapse probability for that environment-condition pair — the fraction of runs that failed. A value of 1.0 means 100% of runs collapsed; a value of 0.0 means 0% collapsed (all survived); values in between (e.g. 0.6, 0.8) mean partial failure across seeds; and a dash means that combination wasn't tested.

R0 (no regularizer): 50/50 runs collapsed. With prediction loss alone, this minimal architecture has no surviving solution in any environment, under either training protocol — a complete, universal failure.

R2 (faithful EP-SIGReg): 33/33 main-factorial runs collapsed on the OU and both point-mass environments (plus 4/5 under the P0.5 control on OU), while surviving 10/10 runs on Lorenz. So R2's failure is environment-dependent and close to binary — it either always fails or always survives, with almost no middle ground.

R1 (moment proxy): survived almost everywhere — under P0 it never collapsed, and even in its worst cells the collapse rate was partial rather than total (60% on OU/P1, 80% on the stable-linear/P1 condition).

A few positives from the paper

Regularizer exposure is equalized across protocols: the statistic is computed per time point on the full batch and averaged over the time points visible to that protocol's loss, never on a flattened pool of temporally correlated frames. Under the main design R1/R2 apply only to encoded latents, never to rolled-out predictions.

Weight selection. λ was tuned per (regularizer, environment) on $\{0.01, 0.03, 0.1, 0.3, 1.0\}$, seed 0, by validation one-step MSE *subject to non-collapse diagnostics*, never on rollout metrics, and frozen across protocols, so tuning cannot manufacture a regularizer–protocol interaction. When every grid point failed the collapse screen, the selector fell back to best validation MSE; this fallback and its consequences are audited in Section 4.1 and Appendix C.

1. The difference between all the environments and their training protocols would have made the effect of regularizer different on each of them . However the setup keeps in mind this discrepancy and tries to equalize the effect .

2. the λ parameter was picked on the basis of whichever candidate had the best one-step validation prediction error, as long as it also passed a basic "did this collapse" sanity check. It wasn't based on any long horizon drift to keep the testing fair .

3. Honest acceptance of the failed experiment rather than trying to come up for alternative explanations and performing a rightful analysis of a negative result by trying to get most out of it

The λ audit. The selection protocol chose λ=1.0 (the grid boundary) for R2 on OU and Lorenz; on both point-mass environments *every* R2 grid point failed the pilot collapse screen, so the fallback selected by validation MSE alone, which under collapse favors the most degenerate solution (validation one-step MSE of a collapsed code is spuriously tiny: 4×10^{-7} on OU). The honest statement is not "EP-SIGReg collapses world models" but: *at weights chosen by a standard validation protocol, the faithful statistic produced degenerate latents on three of four environments, and the failure is invisible, indeed attractive, to one-step validation error*. Whether other weights avoid this is unresolved here (the λ-sensitivity arm was dropped; Section 5).

A candidate mechanism. At exact collapse ($Z \equiv 0$), every projection is degenerate and Eq. (1) evaluates to $1 + {}^{1}\!/\!\sqrt{3} - \sqrt{2} \approx 0.1631$ at γ=${}^{1}\!/\!{}_{2}$: a stationary point of the regularizer with a small, *bounded* value (the statistic is an MMD, bounded by construction). The observed final regularizer

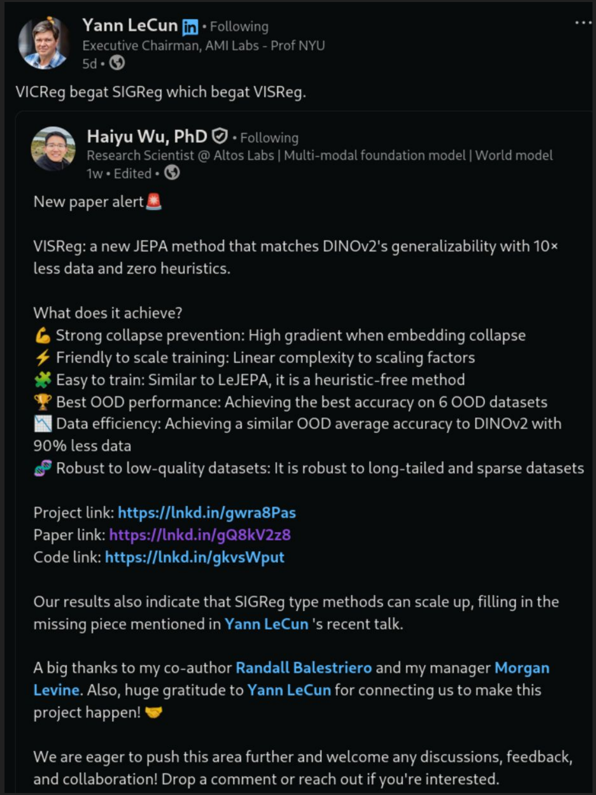
The final conclusion from this run

No concrete new finding - Like my last run , this one even with a higher compute budget couldn't come to a definite conclusion . This can be attributed to the fact that most of the states in the experiment collapsed and often times the learning scope out of negative results gets limited . Even the results that have been mentioned are very tough to comprehend , hence my next steps would be to figure out why those cases collapsed , my most probable guess is that it was a typical representation collapse case that the world models face , but will have to read more on it to be sure .

My suggestions to the Auto Research System

1. In my experience , SOTA models like Claude Fable , Opus 4.8 or GPT 5.5 are way superior to the normal models in the planning phase of the auto-research , and it genuinely brings out the best out of the system . However , these models are very expensive and if we do the complete run on them it would be upwards of 30$ per each run . Hence its not possible to complete the entire session on a model like them.
2. An alternate solution to this that I tried was changing the model to an affordable one like Sonnet post the planning phase . The coding and tracking the experiment part doesn’t actually require Opus or GPT 5.5, and can easily be carried out by inferior models , but changing a model mid- experiment causes the context to lose and this heavily impacts the quality of work .
3. Hence , I would work on a multi-agentic auto-voila where in a Fable and Sonnet work in sync , wherein fable does the planning & writing of the paper and it instructs sonnet, who solely works on the execution part , leaving analysis , writing to a superior model and itself being responsible just for coding and monitoring the GPU runs .
4. According to my estimates , in a 8 hour autoreseatch run , the requirement of model like Fable/Opus is just 45 minutes and rest of the tasks can easily be carried out by Sonnet , hence saving a lot of credits , that can be used on some other work.

Ending Note - Thank you for your time , let me know if there is anything on this that you'd like to know or discuss further !


Yann LeCun • Following
Executive Chairman, AMI Labs - Prof NYU
5d •

VICReg begat SIGReg which begat VISReg.

Haiyu Wu, PhD • Following
Research Scientist @ Altos Labs | Multi-modal foundation model | World model
1w • Edited •

New paper alert🚨

VISReg: a new JEPA method that matches DINOv2's generalizability with 10× less data and zero heuristics.

What does it achieve?
💪 Strong collapse prevention: High gradient when embedding collapse
⚡ Friendly to scale training: Linear complexity to scaling factors
🧩 Easy to train: Similar to LeJEPA, it is a heuristic-free method
🏆 Best OOD performance: Achieving the best accuracy on 6 OOD datasets
📉 Data efficiency: Achieving a similar OOD average accuracy to DINOv2 with 90% less data
🧬 Robust to low-quality datasets: It is robust to long-tailed and sparse datasets

Project link: https://lnkd.in/gwra8Pas
Paper link: https://lnkd.in/gQ8kV2z8
Code link: https://lnkd.in/gkvsWput

Our results also indicate that SIGReg type methods can scale up, filling in the missing piece mentioned in Yann LeCun 's recent talk.

A big thanks to my co-author Randall Balestriero and my manager Morgan Levine. Also, huge gratitude to Yann LeCun for connecting us to make this project happen! 🤝

We are eager to push this area further and welcome any discussions, feedback, and collaboration! Drop a comment or reach out if you're interested.

A sign that regularizer is not just another topic , but a very important one that will be an active area of research in the future. We can already see exciting work coming on this topic like this one that just got released last week !!!

Thank you :)